

Continuación final

1) $x^4(y+7)^3 - 2450 = \tan\left(\frac{x}{y}\right)$ $x=2$ $y=3$

$$x^4 3(y+7)^2 y' + 4x^3(y+7)^3 = \sec^2\left(\frac{x}{y}\right) \left(\frac{y - y'x}{y^2}\right)$$

$$3x^4(y+7)^2 y' + 4x^3(y+7)^3 = \frac{\sec^2\left(\frac{x}{y}\right)(y - xy')}{y^2}$$

$$3(2)^4(3+7)^2 y' + 4(2)^3(3+7)^3 = \frac{\sec^2\left(\frac{2}{3}\right)(3 - 2y')}{3^2}$$

$$4,800 y' + 32,000 = \frac{1.61912(3 - 2y')}{32}$$

$$y' = -6.66649 \approx -6.67$$

2) $\lim_{x \rightarrow 0} (\sinh x)^{\tan x} = (\sinh \cdot 0)^{\tan(0)} = 0^0$

$$\begin{aligned} \ln \lim_{x \rightarrow 0} (\sinh x)^{\tan x} &= \lim_{x \rightarrow 0} \tan(x) \ln(\sinh x) = \\ &= \lim_{x \rightarrow 0} \tan(x) \ln(\sinh \cdot 0) = 0 \cdot \infty \\ &= \lim_{x \rightarrow 0} \frac{\ln(\sinh x)}{\frac{1}{\tan(x)}} = \frac{\ln(\sinh \cdot 0)}{\cot(0)} = \frac{\infty}{\infty} \\ &= \frac{1}{\sinh x} \cdot \frac{1}{\csc^2(x)} = \frac{0}{0} \end{aligned}$$

$$= \frac{h(-\sin(hx)h) \sin(hx) - h \cos(hx) \cos(hx)h}{\sin(hx)^2}$$

$$= \frac{2 \cos(x)}{\sin(x)^3}$$

$$= \frac{h(0)h)0 - h(1)h}{0} = \emptyset$$

$$\ln \lim_{x \rightarrow 0} (\sin hx)^{\tan x} = 0$$

$$\lim_{x \rightarrow 0} (\sin hx)^{\tan x} = e^0 = 1$$

$$3) f(x) = ax^2 + bx + c$$

$$f'(x) = 2ax + b = 0$$

$$x = -\frac{b}{2a} = 2$$

$$-b = 4a \quad (2)$$

$$-b = 4\left(-\frac{8}{32}\right)$$

$$b = 1$$

$$\max = (3, 19)$$

$$f'(2) = -8$$

$$2a(2) + b = -8$$

$$40a + b = -8 \quad (2)$$

$$40a - 8a = -8$$

$$32a = -8$$

$$a = -\frac{8}{32} \approx -0.25$$

$$19 = 40\left(-\frac{8}{32}\right) + 1 + c$$

$$19 = -9 + c$$

$$19 + 9 = c$$

$$c = 28$$